

Haifei Zhang

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Profile

I am currently an Associate Professor (Maître de Conférences in French) at [Jean Monnet University](#) in Saint-Étienne, France. I conduct my research in the Data Intelligence team of the [Hubert Curien Laboratory](#) (UMR-CNRS 5516), a joint research unit of Jean Monnet University, CNRS, and the Institution of Optique Graduate School. I teach at [Télécom Saint-Étienne](#), a highly reputed engineering school.

My research primarily focuses on complex data analysis, graph representation learning, and trustworthy AI, with a particular interest in explainable AI (XAI), cautious classifiers, and uncertainty management in machine learning. I am dedicated to developing robust and interpretable AI models that enhance decision-making in high-stakes applications.

Education

University of Technology of Compiègne PhD in Computer Science Dissertation: Explainable Cautious Classifiers Supervisors: Benjamin Quost , Marie-Hélène Masson	November 2020 - November 2023 France
University of Technology of Compiègne Master's degree in Complex Systems	September 2019 - September 2020 France
University of Technology of Compiègne Engineer's degree in Data Mining	September 2017 - September 2020 France
Shanghai University Bachelor's degree in Information Engineering	August 2014 - July 2017 China

Employment

Jean Monnet University Associate Professor	September 2024 - now France
University of Technology of Compiègne Temporary Lecturer and Research Assistant	November 2023 - August 2024 France

Teaching

Jean Monnet University

Deep learning	Spring 2025
Design patterns	Spring 2025
Introduction to Git	Spring 2025
Web development with PHP	Fall 2024
Advanced algorithms	Fall 2024
University of Technology of Compiègne	
Linear algebra basics	Fall 2023, Spring 2024
Data analysis and data Mining	Spring 2022, Spring 2023, Spring 2024
Statistical methods for engineers	Fall 2021, Fall 2022, Fall 2023, Spring 2024

Publications

Journal papers

1. **Haifei Zhang**, and Jinfeng Zhong. “Efficient and Effective Counterfactual Explanations for Random Forests.” *Expert Systems with Applications* 293 (2025): 128661. [[Paper Link](#)]
2. Nguyen, Vu-Linh, **Haifei Zhang**, and Sébastien Destercke. “Credal ensembling in multi-class classification.” *Machine Learning* 114.1 (2025): 1-62. [[Paper Link](#)]
3. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Cautious classifier ensembles for set-valued decision-making.” *International Journal of Approximate Reasoning* 177 (2025): 109328. [[Paper Link](#)]
4. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Cautious weighted random forests.” *Expert Systems with Applications* 213 (2023): 118883. [[Paper Link](#)]

International conference papers

1. **Zhang, Haifei**. “SHADED: Shapley Value-Based Deceptive Evidence Detection in Belief Functions.” *International Conference on Belief Functions*. Cham: Springer Nature Switzerland, 2024. [[Paper Link](#)]
2. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Cautious Decision-Making for Tree Ensembles.” *European Conference on Symbolic and Quantitative Approaches with Uncertainty*. Cham: Springer Nature Switzerland, 2023. [[Paper Link](#)]
3. Nguyen, Vu-Linh, **Haifei Zhang**, and Sébastien Destercke. “Learning Sets of Probabilities Through Ensemble Methods.” *European Conference on Symbolic and Quantitative Approaches with Uncertainty*. Cham: Springer Nature Switzerland, 2023. [[Paper Link](#)]
4. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Explaining cautious random forests via counterfactuals.” *International Conference on Soft Methods in Probability and Statistics*. Cham: Springer International Publishing, 2022. [[Paper Link](#)]
5. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Cautious random forests: a new decision strategy and some experiments.” *International Symposium on Imprecise Probability: Theories and Applications*. PMLR, 2021. [[Paper Link](#)]

National conference papers

1. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Explications contrefactuelles pour les forêts aléatoires prudentes.” *31èmes Rencontres Francophones sur la Logique Floue et ses Applications (LFA 2022)*. 2022. [[Paper Link](#)]
2. **Zhang, Haifei**, Benjamin Quost, and Marie-Hélène Masson. “Forêts aléatoires prudentes: une nouvelle stratégie de décision et quelques expériences.” *31èmes Rencontres Francophones sur la Logique Floue et ses Applications (LFA 2022)*. 2022. [[Paper Link](#)]

Review service

ECAI	2024, 2025
Knowledge-Based Systems	2025
Forum for Economic and Financial Studies	2025